

if:

It is a decision making statement.

It is used to check the given condition / expression is true or false.

If condition true then only the statements in if block { } are executed. If condition false the if block statements are not executed.

Note: in C other than 0 anything is 1 i.e. true.

C comes with 4 type of if conditions.

1. Simple if – one condition
2. If..else – two
3. If..else if ladder – more than two
4. Nested if

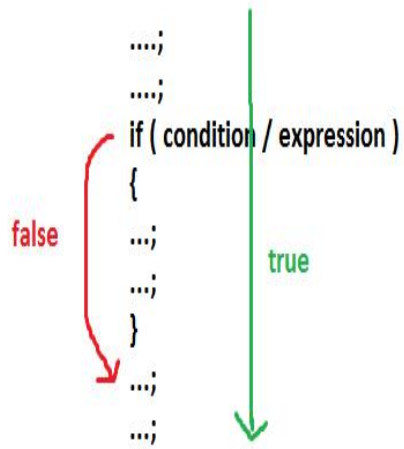
Simple if:

When the program is having only one condition then only go for simple if.

If condition true statements in if block {} are executed and later outside statements also executed.

If condition false only the outside statements are executed.

simple if:



if(condition)
statement;

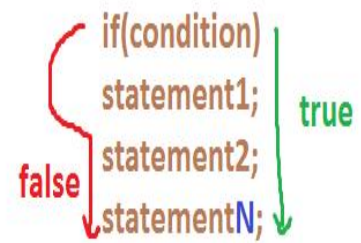
or

```
if( condition )  
{  
  statement;  
}
```

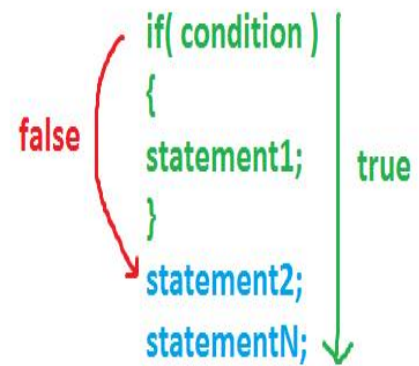
or

if(condition) statement;

or



or



```
TC
File Edit Run Compile Project Options Debug
Error: Expression syntax in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( )_
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* Error */
```

```
TC
File Edit Run Compile Project Options Debug
Error: Expression syntax in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( if( 3>=3 ) )
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* Error */
```

```
TC
File Edit Run Compile Project Options Debug
Error: Division by zero in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( 5%0 )
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* Error */
```

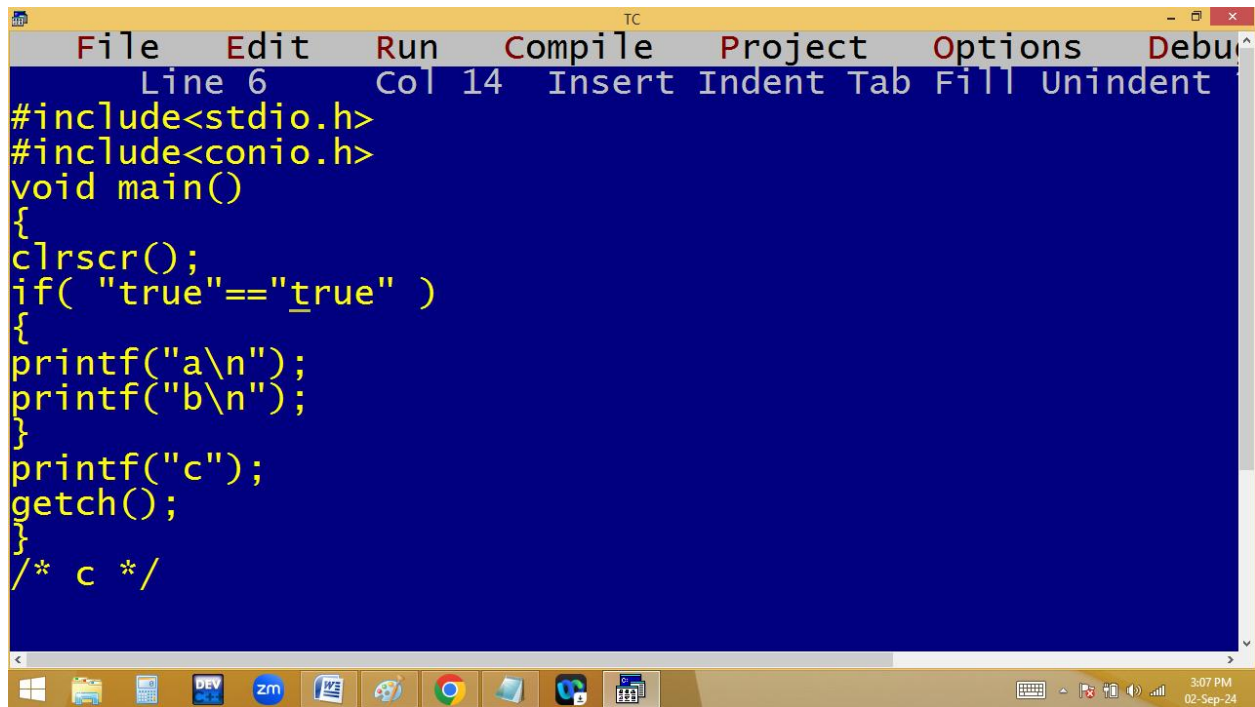
```
TC
File Edit Run Compile Project Options Debug
Error: Illegal use of floating point in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( 5.0%2 )
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* Error */
```

```
TC
File Edit Run Compile Project Options Debug
Error: Undefined symbol 'true' in function main
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( true )
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* Error */
```

```
TC
File Edit Run Compile Project Options Debug
Line 15 Col 52 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( "false" )
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* abc
"false" is a string which contains non-zero address_
*/
```

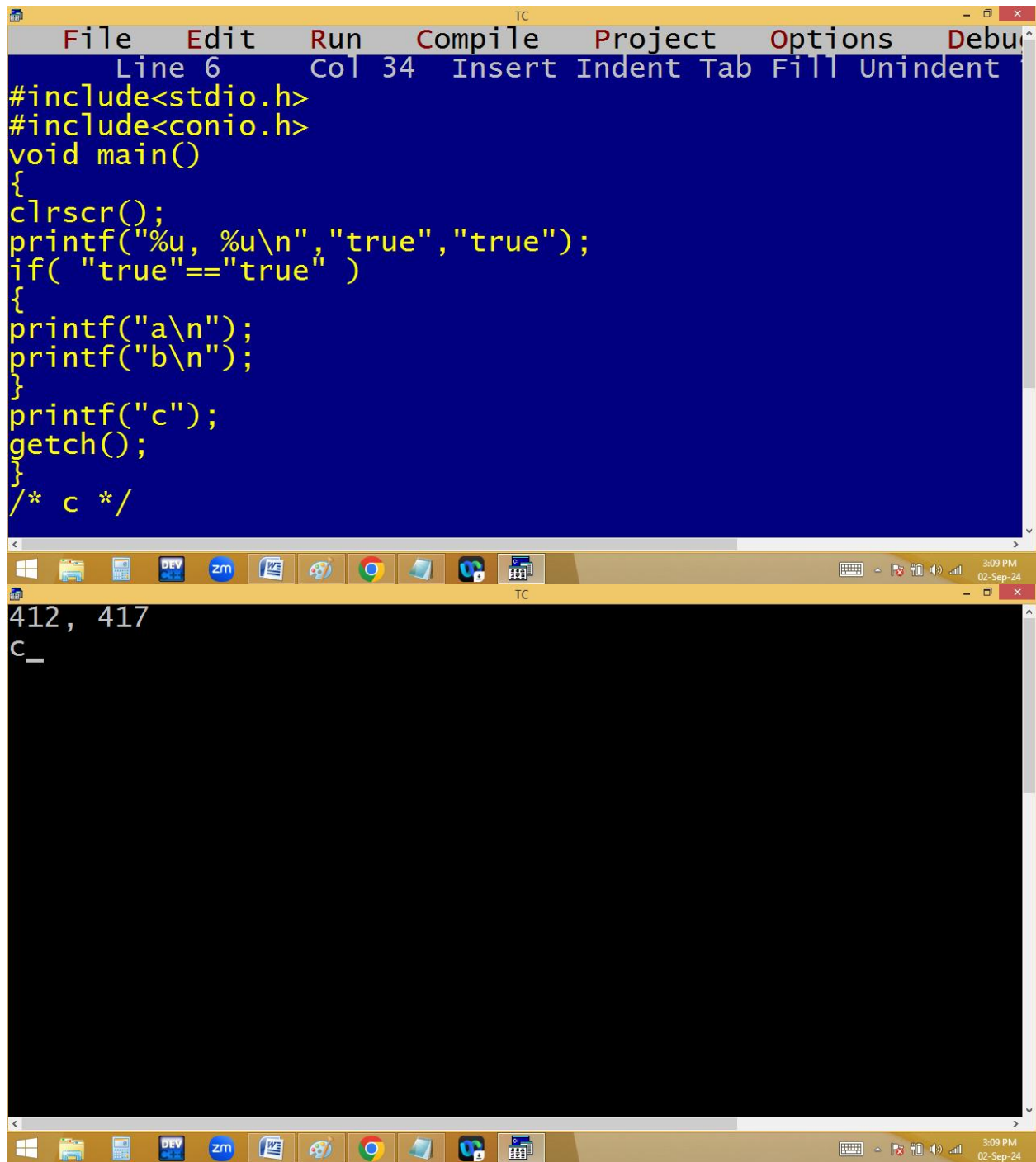
```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 8 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( "true" - "false" )
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* abc */
```

```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 4 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( "true" > "false" )
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* c */
```

```
TC
File Edit Run Compile Project Options Debug
Line 6 Col 14 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( "true"=="_true" )
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* c */
```

Windows taskbar at the bottom shows icons for File Explorer, DEV, zm, VS Code, Chrome, and others. The system tray on the right shows the time as 3:07 PM on 02-Sep-24.



The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays a C program with the following code:

```
File Edit Run Compile Project Options Debug
Line 6 Col 34 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%u, %u\n", "true", "true");
if( "true"=="true" )
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* c */
```

The bottom window shows the output of the program:

```
412, 417
c_
```

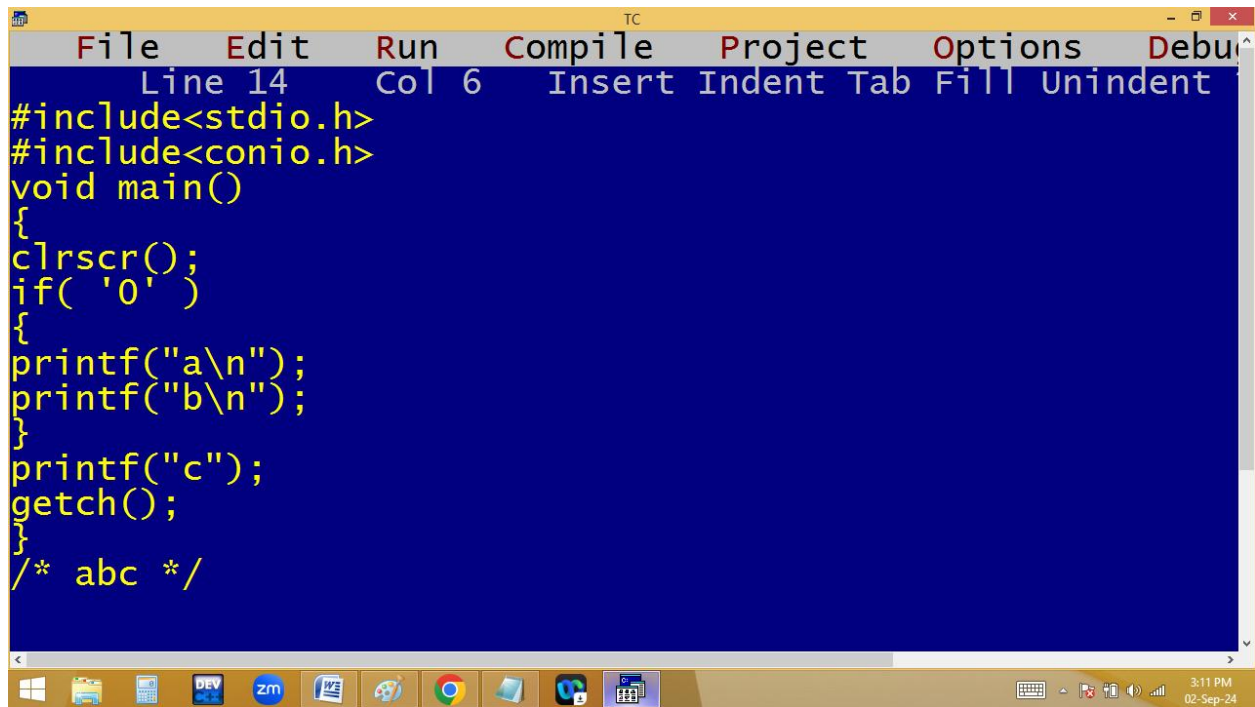
The Windows taskbar at the bottom indicates the time is 3:09 PM on 02-Sep-24. The taskbar includes icons for Windows, File Explorer, DEV, zm, WE, a paint application, Google Chrome, and the Turbo C++ IDE.

The image shows a screenshot of the Turbo C++ (TC) IDE. The top window is the editor, which has a blue background and contains the following C code:

```
File Edit Run Compile Project Options Debug
Line 6 Col 9 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( 'a'/'b' )
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* c */
```

The bottom window is the output console, which has a black background and displays the character 'C'.

The Windows taskbar is visible at the bottom of the screen, showing the Start button, several application icons (including DEV, zm, and Chrome), and the system clock indicating 3:10 PM on 02-Sep-24.

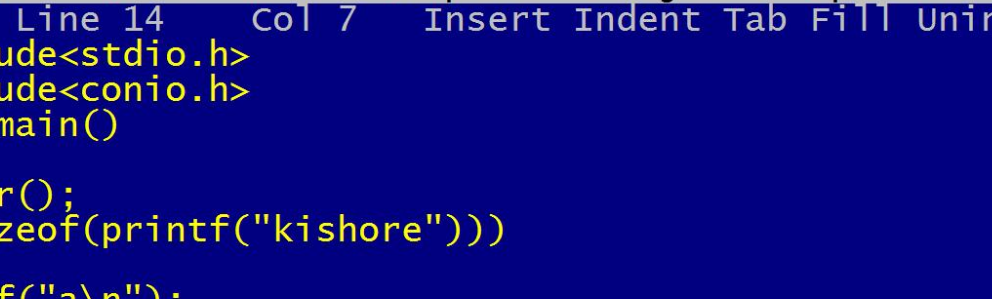


```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 6 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( '0' )
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* abc */
```

Windows taskbar icons: File Explorer, DEV, zm, Word, Paint, Chrome, Edge, Task Manager. System tray: 3:11 PM, 02-Sep-24.

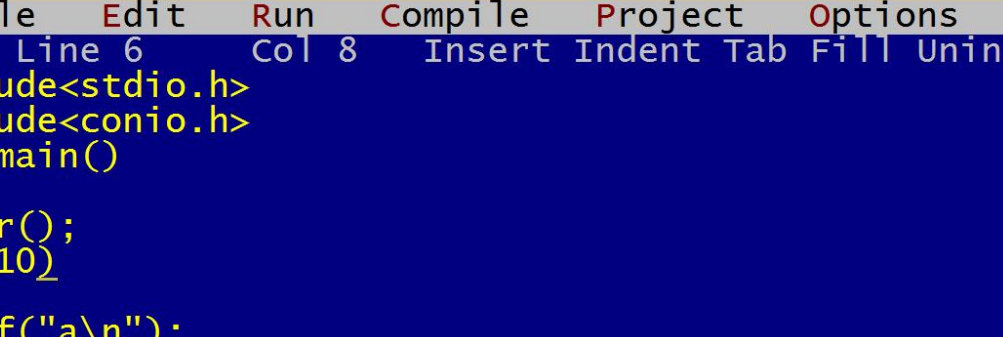
```
TC
File Edit Run Compile Project Options Debug
Line 6 Col 9 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( '0'>0 )
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* abc */
```

```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 11 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if( printf("kishore")>sizeof("kishore"))
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* Kishorec */
```



The screenshot shows the Turbo C++ IDE interface. The menu bar at the top includes File, Edit, Run, Compile, Project, Options, and Debug. Below the menu bar, the status bar indicates 'Line 14 Col 7' and lists keyboard shortcuts: Insert, Indent, Tab, Fill, and Unindent. The main editing area has a dark blue background with yellow text. The code is as follows:

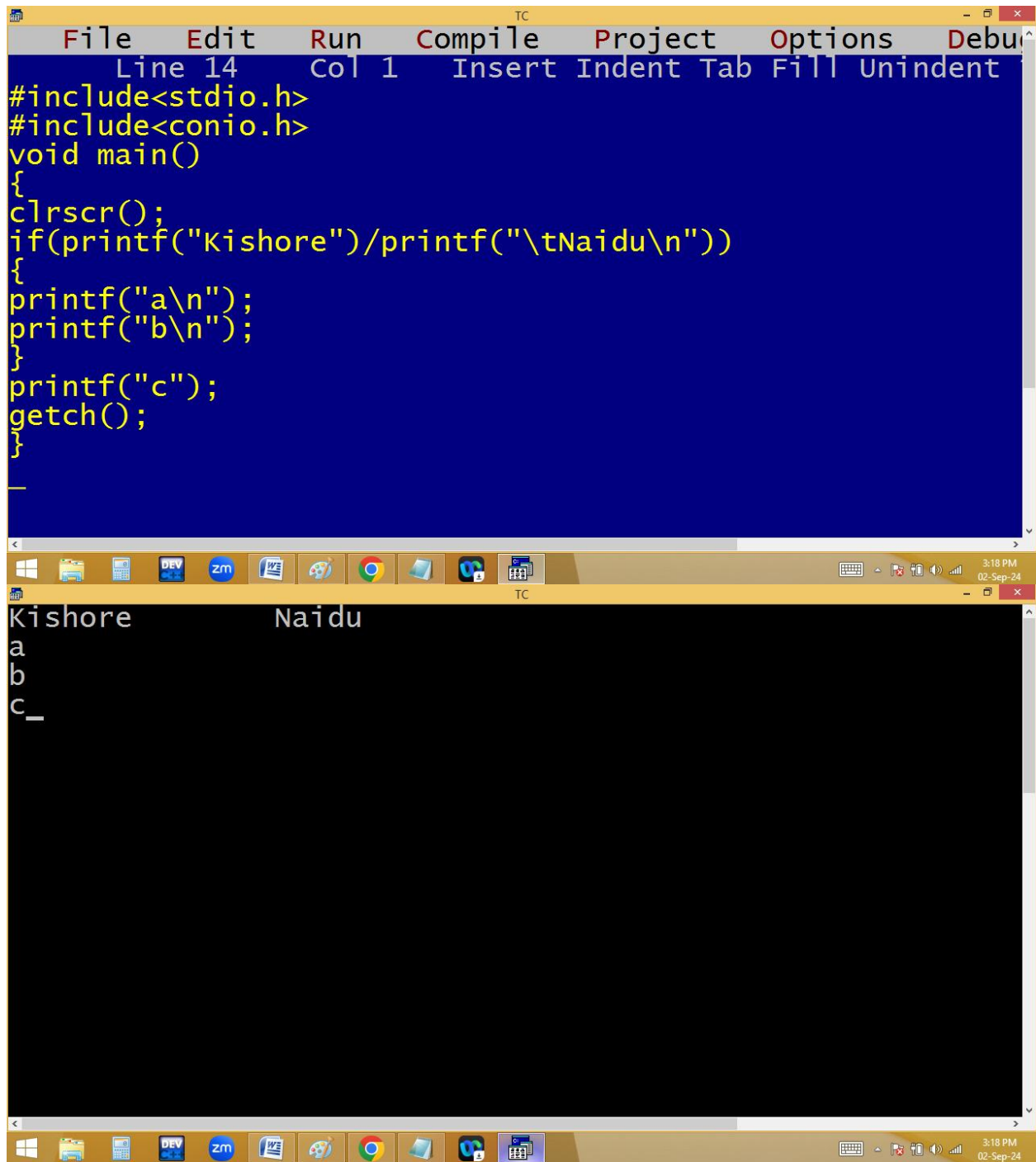
```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(sizeof(printf("kishore")))
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* abc */
```



The screenshot shows the Turbo C++ IDE interface. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 6' and 'Col 8'. The code editor contains the following C program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(0.10)
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
/* abc */
```

The program is a simple C program that includes `stdio.h` and `conio.h`. It defines a `main` function that calls `clrscr()` to clear the screen, then checks a condition `if(0.10)`. Inside the `if` block, it prints 'a' and 'b' on separate lines. After the `if` block, it prints 'c' on the same line. Finally, it calls `getch()` to wait for a key press before exiting. A comment `/* abc */` is at the bottom of the code.



The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays a C program with the following code:

```
File Edit Run Compile Project Options Debug
Line 14 Col 1 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(printf("Kishore")/printf("\tNaidu\n"))
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}
```

The bottom window shows the output of the program:

```
Kishore      Naidu
a
b
c_
```

The Windows taskbar at the bottom shows the time as 3:18 PM on 02-Sep-24. The taskbar includes icons for Windows, File Explorer, DEV, zm, Word, Paint, Chrome, and the TC application.

```
TC
File Edit Run Compile Project Options Debug
Line 15 Col 5 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(0);
{
printf("a\n");
printf("b\n");
}
printf("c");
getch();
}

/* abc */
```

```
TC
File Edit Run Compile Project Options Debug
Line 15 Col 16 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(0)
{
printf("a\n");
printf("b\n");
printf("c");
}
getch();
}

/* blank screen_*/
```


The image displays two instances of the Turbo C++ (TC) IDE, stacked vertically. Both windows show the same C program code, which is designed to print the characters 'a', 'b', and 'c' on separate lines. The code includes the `stdio.h` and `conio.h` headers, defines a `main` function, and uses `clrscr()` to clear the screen. It then uses `printf` to output 'a\n', 'b\n', and 'c', followed by `getch()` to pause the execution. The code is enclosed in a multi-line comment `/* bc_ */`.

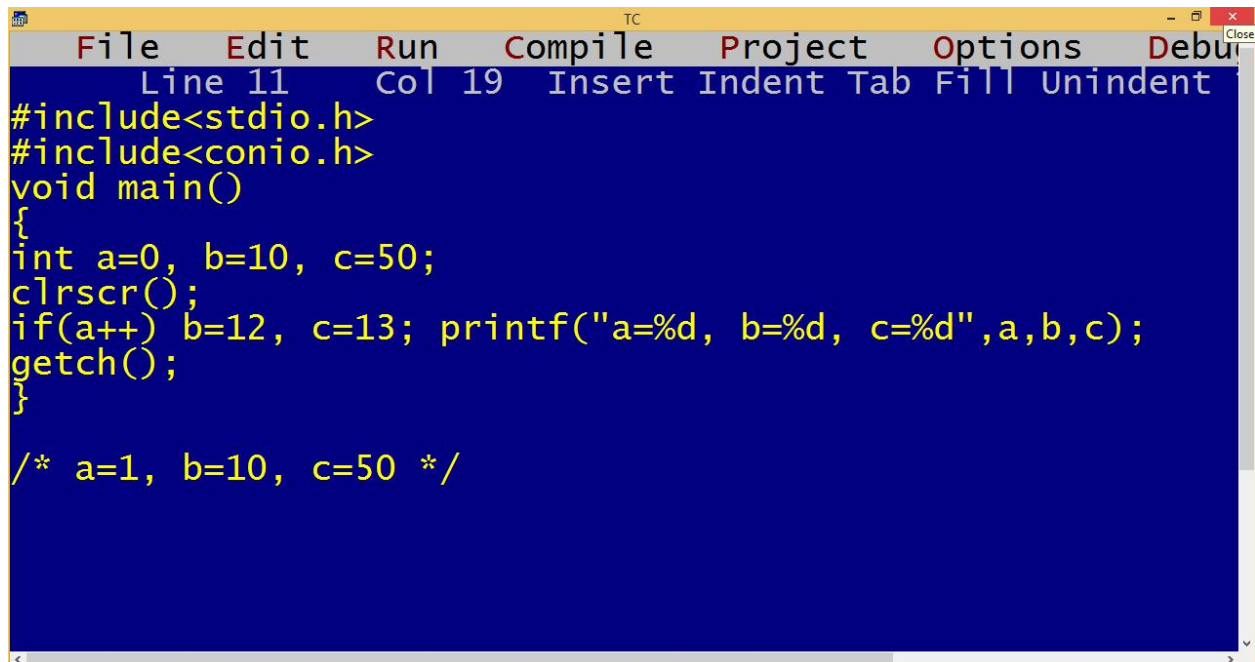
The top window shows the cursor at Line 13, Column 6. The bottom window shows the cursor at Line 9, Column 2. Both windows have a menu bar with 'File', 'Edit', 'Run', 'Compile', 'Project', 'Options', and 'Debug'. The Windows taskbar at the bottom of each window shows various application icons and the system clock indicating 3:21 PM on 02-Sep-24.

```
File Edit Run Compile Project Options Debug
Line 13 Col 6 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(0)
printf("a\n");
printf("b\n");
printf("c");
getch();
}

/* bc_ */
```

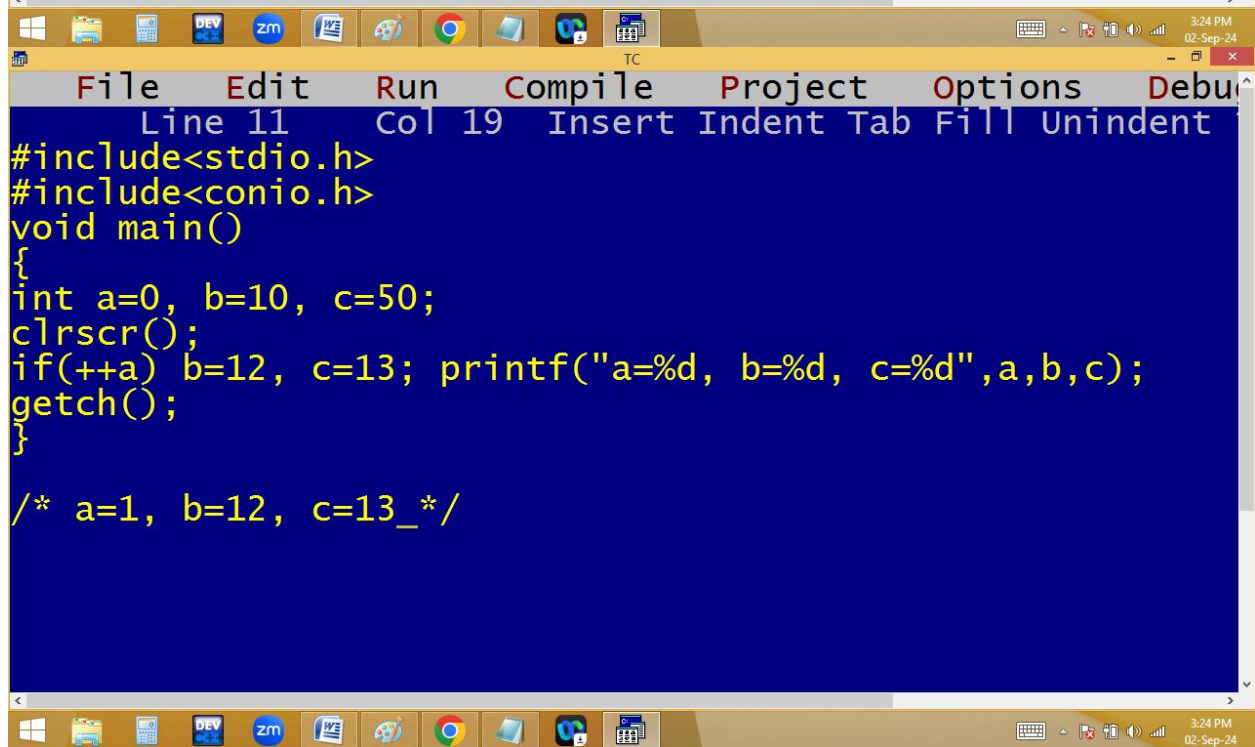
```
File Edit Run Compile Project Options Debug
Line 9 Col 2 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(0)
{
printf("a\n");
}_
printf("b\n");
printf("c");
getch();
}

/* bc */
```



```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 19 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0, b=10, c=50;
clrscr();
if(a++) b=12, c=13; printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}

/* a=1, b=10, c=50 */
```



```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 19 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0, b=10, c=50;
clrscr();
if(++a) b=12, c=13; printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}

/* a=1, b=12, c=13_*/
```

```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 19 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a=9, b=10, c=50;
clrscr();
if(a=0) b=12, c=13; printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}

/* a=0, b=10, c=50 */
```

```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 19 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a=9, b=10, c=50;
clrscr();
if(a=0,a=1) b=12, c=13; printf("a=%d, b=%d, c=%d",a,b,c);
getch();
}

/* a=1, b=12, c=13_*/
```

```
TC
File Edit Run Compile Project Options Debug
Line 11 Col 7 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int a=9, b=10, c=50;
clrscr();
if(a=0,a=1,a--) b=12, c=13; printf("a=%d, b=%d, c=%d",a,b
getch();
}

/* a=0, b=12, c=13 */
```

```
TC
File Edit Run Compile Project Options Debug
Line 8 Col 2 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(printf("kishore"))
{
}
getch();
}
```

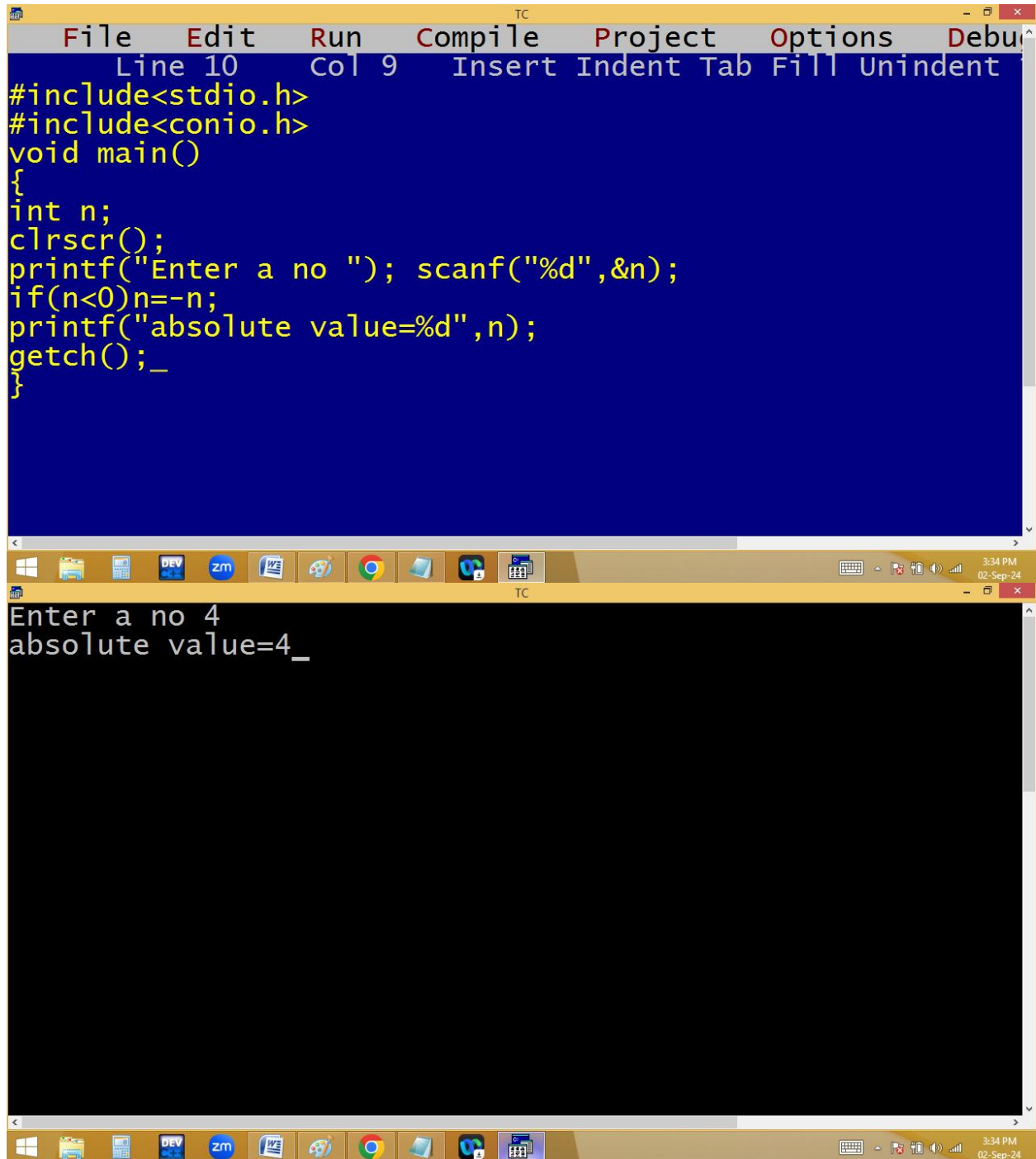
The screenshot shows the Turbo C++ (TC) IDE interface. The top window, titled 'TC', is a black console window displaying the text 'kishore'. Below it is the main editor window, also titled 'TC', which has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
if(printf("kishore"))
}_
```

A red error message banner is visible at the top of the editor window, stating: "Error: Statement missing ; in function main". The code appears to be a program that prints 'kishore' and then calls `clrscr()` to clear the screen. The IDE's menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The Windows taskbar at the bottom shows the time as 3:29 PM and 3:30 PM on 02-Sep-24.

Finding absolute value of given no[always +Ve]:

If($n < 0$) $n = -n$;

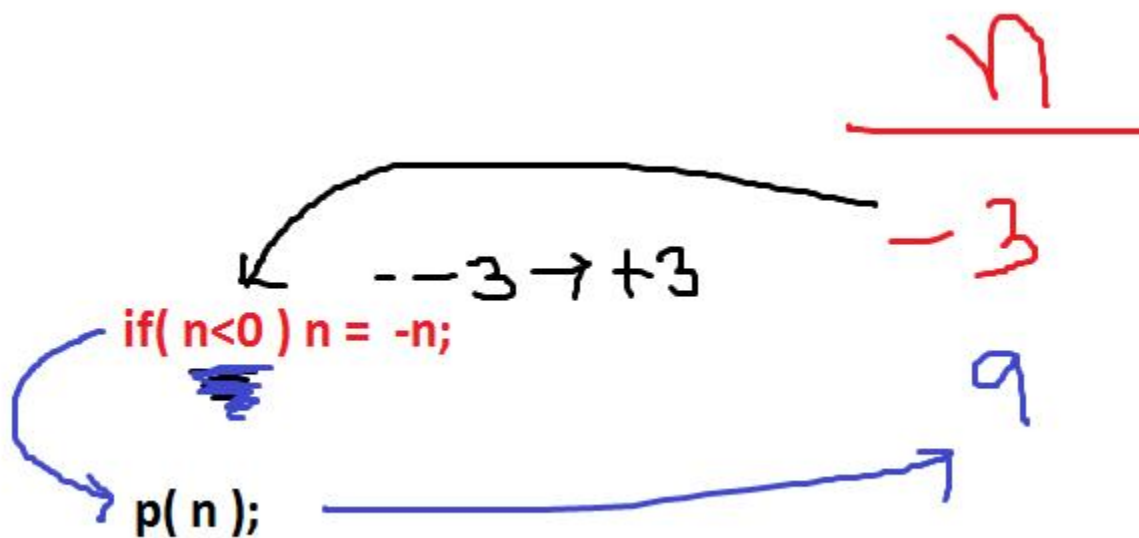


The screenshot displays the Turbo C++ (TC) IDE. The top window shows the source code for a program that calculates the absolute value of a user-input number. The code includes `<stdio.h>` and `<conio.h>`, uses `clrscr()` to clear the screen, and employs `scanf` to read an integer `n`. It then uses an `if` statement to check if `n` is negative; if so, it negates `n` (`n = -n`). Finally, it prints the absolute value using `printf` and waits for a key press with `getch()`.

```
File Edit Run Compile Project Options Debug
Line 10 Col 9 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter a no "); scanf("%d",&n);
if(n<0)n=-n;
printf("absolute value=%d",n);
getch();_
}
```

The bottom window shows the program's execution. It prompts the user to "Enter a no" and the input "4" is provided. The output is "absolute value=4_". The Windows taskbar at the bottom indicates the time is 3:34 PM on 02-Sep-24.


```
TC
Enter a no -9
absolute value=9
```

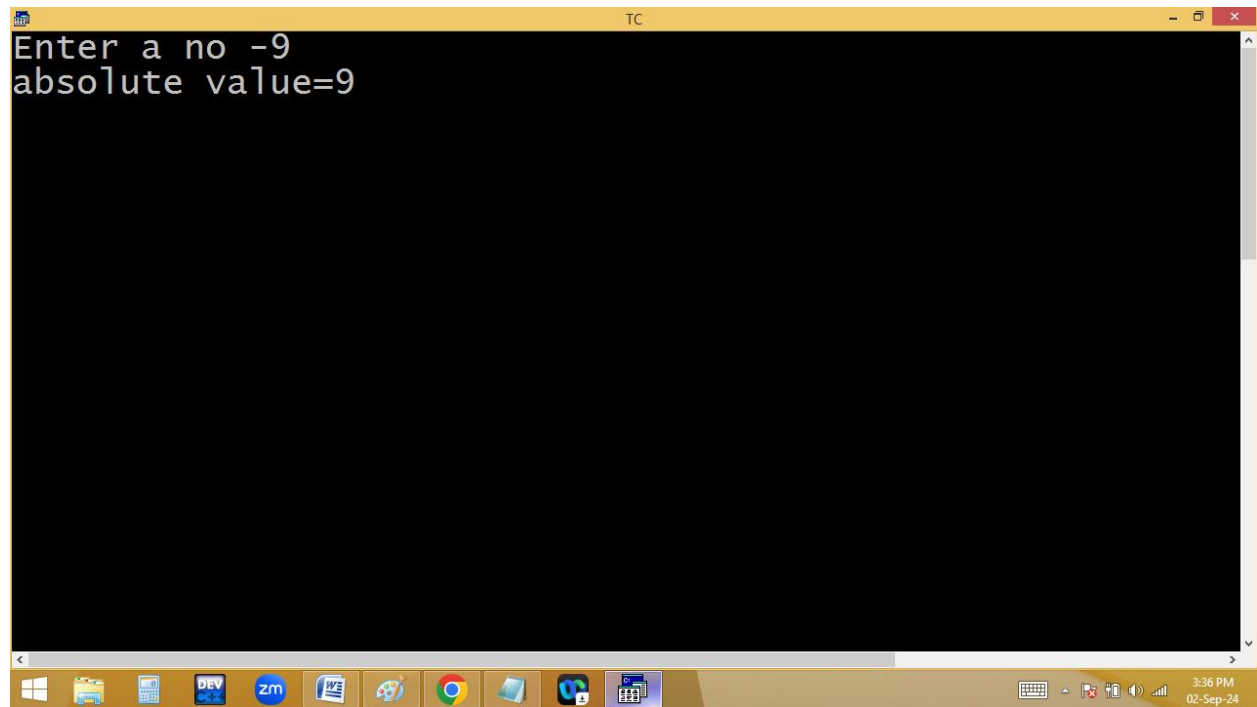


Using abs():

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code of a C program designed to calculate the absolute value of a user-entered integer. The code includes headers for standard input/output, console I/O, and math functions. The main function prompts the user to enter a number, reads the input, and prints the absolute value using the `abs()` function. The bottom screenshot shows the program's execution. It prompts the user to "Enter a no" and shows the input "3". The output is "absolute value=3".

```
File Edit Run Compile Project Options Debug
Line 3 Col 17 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
int n;
clrscr();
printf("Enter a no "); scanf("%d",&n);
printf("absolute value=%d",abs(n));
getch();
}
```

Enter a no 3
absolute value=3



The screenshot shows a Turbo C++ (TC) window with a black background. The text "Enter a no -9" is on the first line, and "absolute value=9" is on the second line. The window title bar says "TC". The Windows taskbar is visible at the bottom with various icons and a system clock showing 3:36 PM on 02-Sep-24.

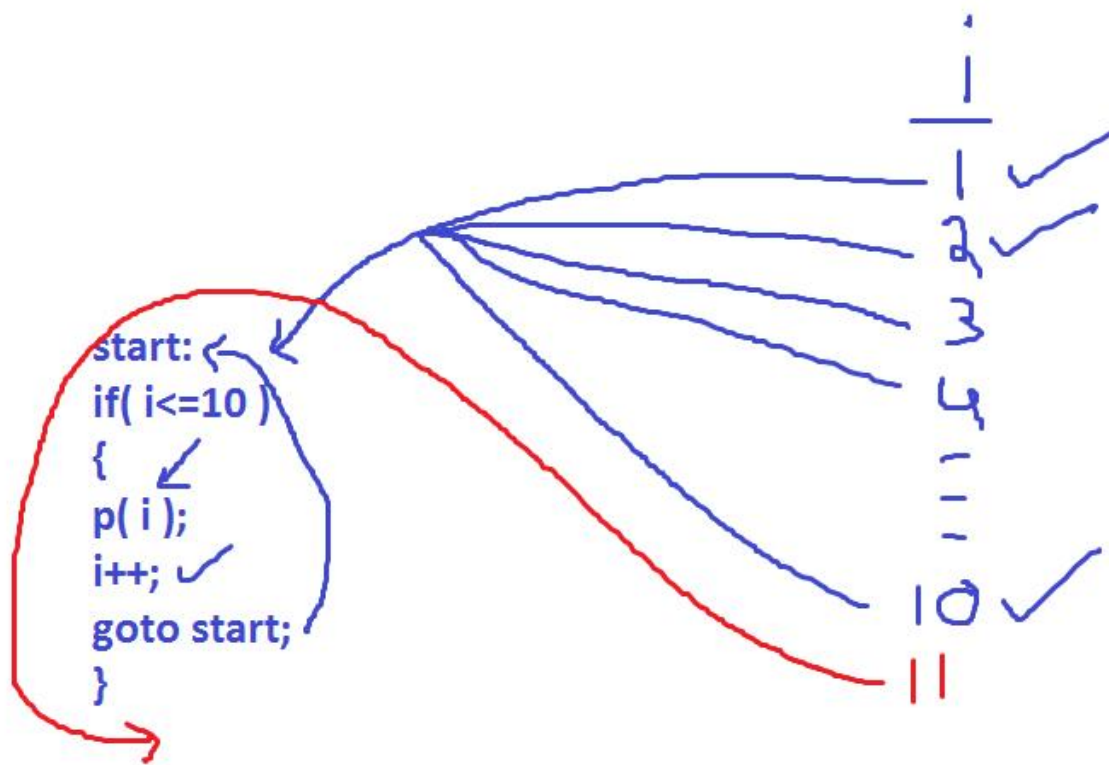
```
Enter a no -9
absolute value=9
```

Printing 1..0 numbers without using loop:

The image shows a screenshot of the Turbo C++ (TC) IDE. The top window is the code editor, which has a blue background and contains the following C code:

```
File Edit Run Compile Project Options Debug
Line 13 Col 2 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1;
clrscr();
start:
if(i<=10)
{
printf("%d\n",i);
i++;
goto start;
}
_getch();
}
```

The bottom window is the output console, which has a black background and displays the numbers 1 through 10, each on a new line, indicating the successful execution of the program.

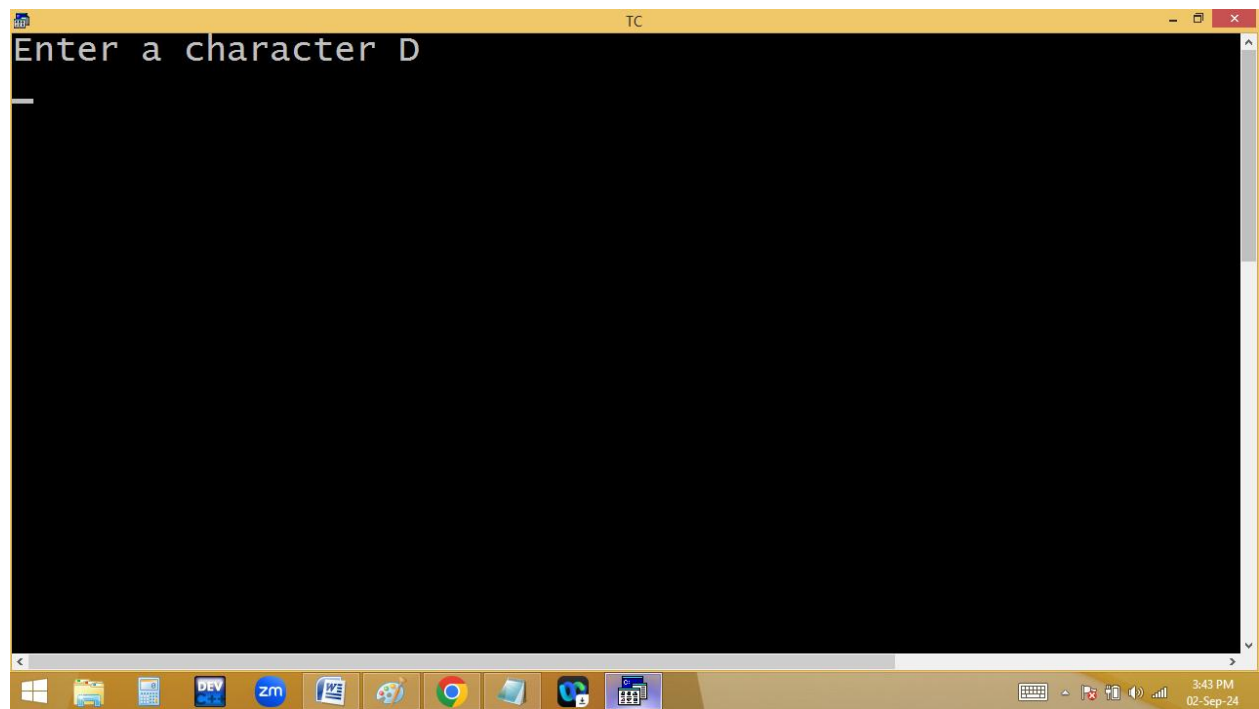


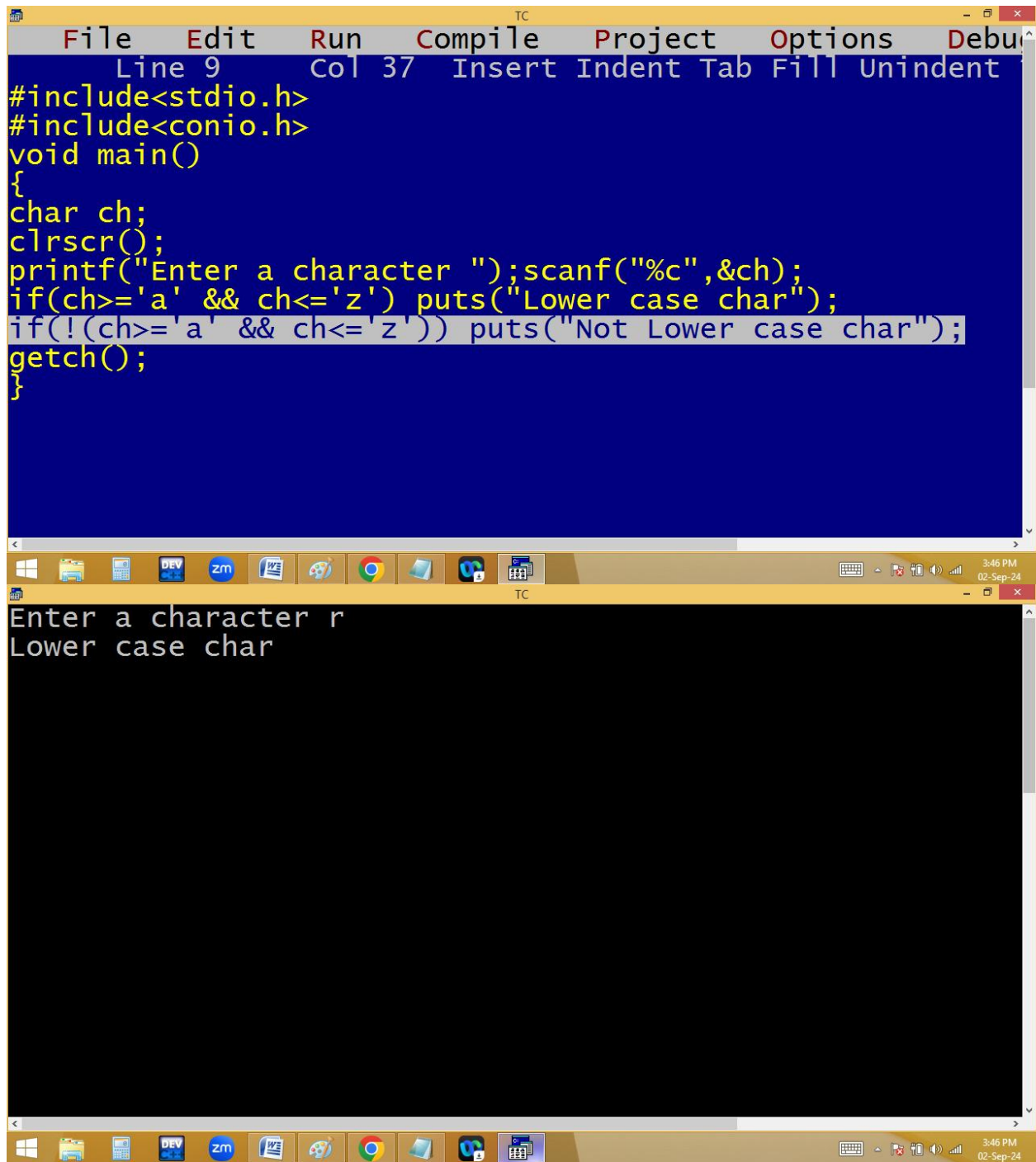
Finding lower case alphabet or not using simple if:

The image shows two windows of the Turbo C++ (TC) IDE. The top window is the editor, displaying a C program with the following code:

```
File Edit Run Compile Project Options Debug
Line 8 Col 48 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter a character ");scanf("%c",&ch);
if(ch>='a' && ch<='z') puts("Lower case char");
getch();
}
```

The bottom window is the output console, showing the program's execution. It displays the prompt "Enter a character f" and the output "Lower case char". A cursor is visible on the line following the output.





The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the source code of a C program. The code is as follows:

```
File Edit Run Compile Project Options Debug
Line 9 Col 37 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter a character ");scanf("%c",&ch);
if(ch>='a' && ch<='z') puts("Lower case char");
if(!(ch>='a' && ch<='z')) puts("Not Lower case char");
getch();
}
```

The bottom window shows the output of the program. It displays the prompt "Enter a character r" and the output "Lower case char".

The screenshot displays the Turbo C++ (TC) IDE interface. The top window shows the program's output, and the bottom window shows the source code.

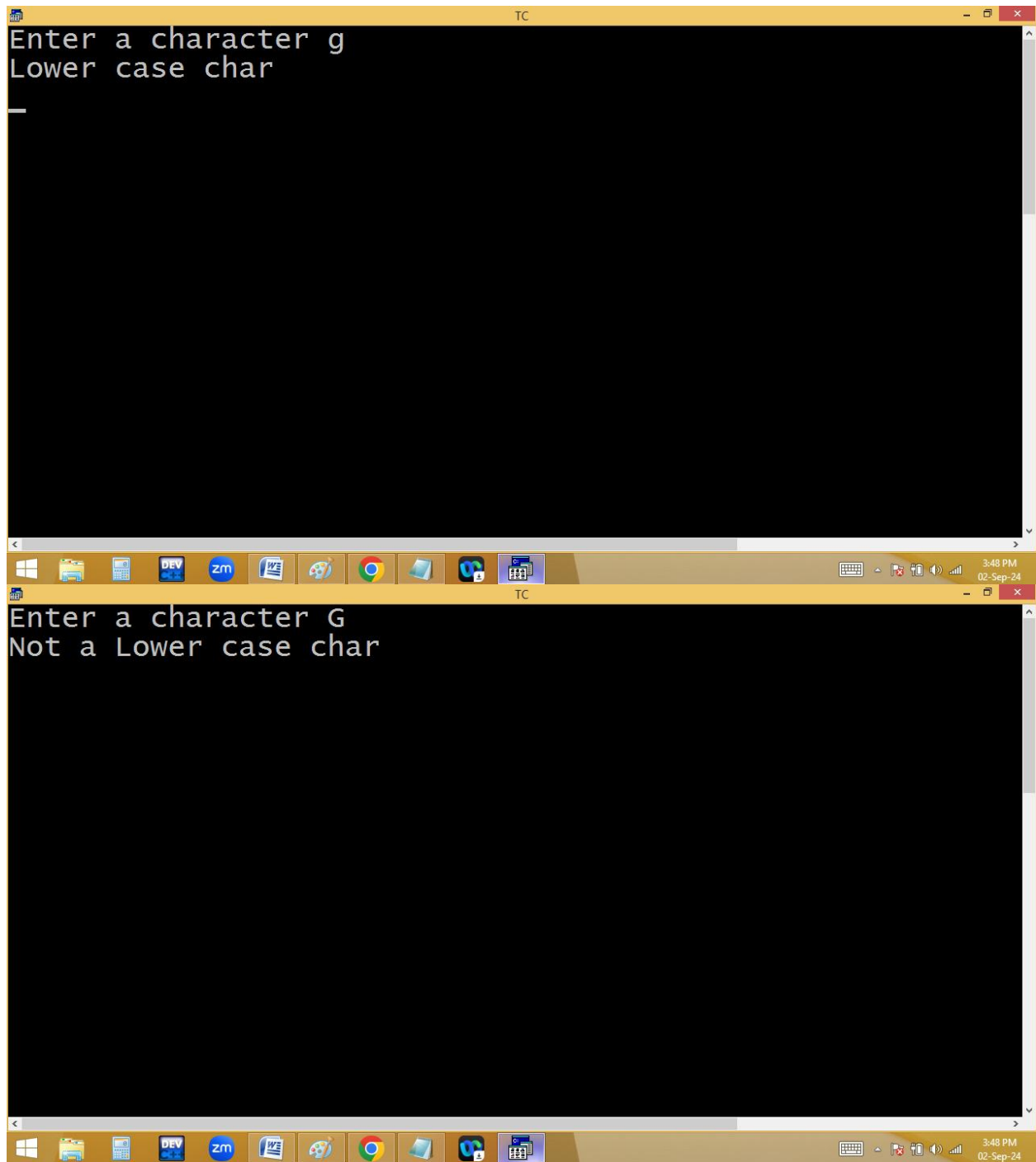
Output Window:

```
Enter a character M
Not Lower case char
```

Source Code Window:

```
File Edit Run Compile Project Options Debug
Line 9 Col 1 Insert Indent Tab Fill Unindent
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter a character ");scanf("%c",&ch);
if(ch>='a' && ch<='z') puts("Lower case char");
if(ch<'a' || ch>'z') puts("Not a Lower case char");
getch();
}
```

The code in the source window contains a syntax error on line 9, which is highlighted in red. The error is in the condition `if(ch<'a' || ch>'z')`, where the single quotes around 'a' and 'z' are not properly closed, resulting in a compiler error.



Finding special character or not?

The image shows a screenshot of the Turbo C++ (TC) IDE. The top window is the code editor, which has a blue background and a yellow menu bar. The menu bar includes 'File', 'Edit', 'Run', 'Compile', 'Project', 'Options', and 'Debug'. The status bar at the bottom of the editor shows 'Line 9', 'Col 5', and a list of actions: 'Insert', 'Indent', 'Tab', 'Fill', 'Unindent'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter a character ");scanf("%c",&ch);
if(ch>='a' && ch<='z') puts("Lower case char");
else puts("Not a Lower case char");
getch();
}
```

The bottom window is the output console, which has a black background. It displays the output of the program: 'Enter a character R' followed by 'Not a Lower case char'. The Windows taskbar is visible at the bottom of the screen, showing various application icons and the system clock indicating 3:50 PM on 02-Sep-24.

